

CONTACT	Phone: +55 (48) 991139920 Email: gabifertig08@gmail.com Postal code: 88934-210
EDUCATION	Federal Institute of Santa Catarina (IFSC) Technical High School in Electronics, 2022 – Present
PROFESSIONAL EXPERIENCE	Robotics Team Programmer — FRC5800 Magic Island Robotics - 2022-2024 - Engineered and programmed an industrial-grade competition robot for the First Robotics Competition (FRC), utilizing Java and the WPILib library to manage complex mechanical systems. - Resolved critical programming errors in real-time as the Robot Pilot during high-pressure international competitions, including the Brunswick Eruption Off Season in New Jersey, USA (receiving a merit-based scholarship). - Led the web development efforts to design and implement a mobile scouting application (with a team of 3) using SvelteKit, JavaScript, CSS, and Tailwind, significantly optimizing data collection for strategic match planning. - Demonstrated strong communication and leadership by presenting awards at the "Build Season 2023 - Brasília" event. - Participated in major technology events, including the Latin American Congress of Free Software and Open Technologies (Latinoware 2023). Web Development Intern — VegomTec - July 2023 – December 2023 - Executed front-end corrections and essential adaptations on a large-scale company website (hundreds of pages), improving client interaction and streamlining the service request flow. - Developed the front-end for a mobile business management application, utilizing a stack of HTML, CSS, JavaScript, and Bootstrap. Drone Programmer — IFSC DeltaHigh Drone Competition Team - April 2024 – Present - Develop autonomous programming for quadrotors competing in the Brazilian Robotics Competition (CBR). - Implemented and integrated AI and Machine Learning-driven algorithms using Python and OpenCV to handle diverse tasks, enabling advanced autonomous navigation, object detection, and complex decision-making for competition requirements. - Computer Vision & AI Specialist: Implemented advanced detection algorithms (YOLOv8) for autonomous navigation and served as a guest speaker at the 2025 Baiana Drone Olympiad (OBAdrones), presenting on AI techniques and model training applied to competition drones.

Volunteer Researcher — IFSC Research & Extension Projects - July 2023 – November 2024

- Spearheaded the development and launch of the "Educational Robotics Platform" website, utilizing WordPress to enhance the project's outreach and informational accessibility.
- Conducted research for the "Innovation in Autonomous Competition Robots using Artificial Intelligence" project, applying Python, Computer Vision, and AI techniques to develop advanced autonomous robot capabilities.
- Robotics Teaching & Digital Inclusion | Instituto Padre Vilson Groh (IVG) / IFSC: Served as an instructor in robotics workshops for children and youth in underserved communities. Conducted didactic classes using Arduino and basic electronics, focusing on fostering logical reasoning, creativity, and democratizing access to technology as a tool for social transformation.

Research Scholarship Holder — IFSC Research Project (Embrapii Funded) - Dec 2024 – Dec 2025

- Led the development, implementation, and bench testing of electronic circuits for a medical-grade device designed to measure electrical and sensor parameters in hospital environments.
- Programmed microcontrollers to ensure efficient data collection and processing from developed devices.
- Developing an interactive, intuitive dashboard to display collected data, integrated with Amazon AWS cloud services to ensure secure data storage and availability.

PROFILE

Highly motivated Software and Web Developer with a proven background in embedded systems, robotics, and full-stack web development. Skilled in leveraging Python, JavaScript, and modern frameworks to deliver robust solutions and optimize system performance. Experienced in implementing AI/Computer Vision for autonomous tasks and possessing a foundational interest in Game Development. Fast learner and active contributor.

SKILLS

- **Programming Languages:** Python (Advanced), JavaScript (Advanced), Java (Advanced), C (Advanced), C++ (Intermediate), C# (Basic), PHP (Intermediate).
- **Web Development (Full-Stack):** HTML (Advanced), CSS (Intermediate), Node.js/Express (Advanced), Tailwind (Intermediate), React (Intermediate), SvelteKit (Intermediate), Three.js (Intermediate), jQuery (Intermediate), WordPress (Intermediate).
- **Databases & Backend:** MongoDB (Advanced), Mongoose (Advanced), MySQL (Advanced).
- **Software & Version Control:** Git and GitHub (Advanced), Docker (Basic).
- **Specialized Expertise:**
 - AI/Vision: Computer Vision (OpenCV), Implementation of AI/ML Algorithms (Advanced).
 - MERN systems (Mongoose, Express, React, Node)

LANGUAGE

- Portuguese: Native
- English: Fluent
- Spanish: Intermediate

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